

SECTION A [12*0.75 = 9 Marks]

Write your answers in the table below

Q #	1	2	3	4	5	6	7	8	9	10	11	12
Answer												

1. DES follows:

- A) Feistel Cipher Structure B) SP Networks
C) Hash Algorithm D) Caesars Cipher

2. The Initial Permutation table/matrix in DES is of size

- A) 16×8 B) 12×8 C) 8×8 D) 4×8

3. The number of unique substitution boxes in DES after the 48 bit XOR operation are

- A) 8 B) 4 C) 6 D) 12

4. In the DES algorithm the 64 bit key input is shortened to 56 bits by ignoring every 4th bit.

- A) True B) False

5. The number of tests required to break the DES algorithm are

- A) 2.8×10^{14} B) 4.2×10^9 C) 1.84×10^{19} D) 7.2×10^{16}

6. Which not a valid key for affine cipher?

- A) K=(17,5) B) K=(3,5) C) K=(2,8) D) K=(7,9)

7. A preferable cryptographic algorithm should have a strong avalanche effect.

- A) True B) False

8. An encryption scheme is unconditionally secure if the ciphertext generated does not contain enough information to determine uniquely the corresponding plaintext, no matter how much cipher text is available.

- A) True B) False

9. Monoalphabetic ciphers are stronger than Polyalphabetic ciphers because frequency analysis is tougher on the former.

- A) True B) False

10. Caesar Cipher is an example of

- A) Poly-alphabetic Cipher
- B) Mono-alphabetic Cipher
- C) Multi-alphabetic Cipher
- D) Bi-alphabetic Cipher

11. Suppose for any BLOCK Cipher you decide to use a 10-bit key. How many possible key combinations are there?

- A) 2^{10}
- B) !10
- C) 10
- D) 10^2

12. How long would a 10-bit message be after being encrypted by a stream cipher?

- A) 2 bits
- B) 6 bits
- C) 10 bits
- D) 20 bits

SECTION B [11 Marks]

1. What will be the estimated computations required to crack a password of 6 characters from the 26 letter alphabet? [1 Mark]

Answer: _____

Space for Calculation:

2. What are the design parameters of Triple DES(Data Encryption Standard)? [2 Marks]

Answer:

Block size : _____

Key size : _____

Number of Rounds: _____

3. For DES S1 TABLE is given below.

[3 Marks]

S-box S_1																
S_1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	01	10	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

Show that $S_1(x_1) \oplus S_1(x_2) \neq S_1(x_1 \oplus x_2)$, where “ $\oplus$ ” denotes bitwise XOR and $S_1(x_1)$ = output of the S-box 1 for x_1 ., for:

(i) $x_1 = 000000, x_2 = 000001$ Answer:	Space for Rough Work:
(ii) $x_1 = 111111, x_2 = 100000$ Answer:	Space for Rough Work:
(iii) $x_1 = 101010, x_2 = 010101$ Answer:	Space for Rough Work:

4. In the **Caesar cipher**, we encrypt a letter by **adding** a fixed number (mod 26) and decrypt by **subtracting** the same number (mod 26). Answer following questions on the basis of Caesar Cipher: [2 Marks]

- (i) What if, instead of **adding**, we use **multiplication (mod 26) for encryption**?

Answer:

- (ii) How would you **decrypt** the message in this case?

Answer:

- (iii) What does it mean to "**divide**" in **modular arithmetic (mod 26)**

Answer:

- (iv) Are there any numbers that **won't work** for multiplication-based encryption? Why?

Answer:

5. A **Linear Feedback Shift Register (LFSR)** is defined by the characteristic polynomial:

$$x^4 + x + 1$$

Starting with an **initial seed (state) of 4 bits**, determine the **first 15 bits** of the key sequence generated by this LFSR. Represent the output sequence **clearly**. [3 Marks]

Solution:

LFSR Structure:

15 bit key stream:

[illegible]

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SPACE FOR ROUGH WORK

Decimal Binary Hexadecimal Table



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STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER:

* FINAL EXAMINATION SPRING-2024 *

COURSE NAME:

COMPUTER SECURITY

COURSE NUMBER:

SOC4010

EXAMINATION DATE:

TIME:

EXAMINATION DURATION:

60 minutes

ADDITIONAL MATERIALS
ALLOWED TO USE:

SIMPLE CALCULATOR

SPECIAL INSTRUCTIONS:

- Maximum Marks: 30
- FILL ANSWERS IN THE PROVIDED **TABLES**

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

All Calculations/steps must be shown at the space provided after the question (penalty may be applicable as instructed in class)

Ambiguous answers will be awarded zero. **Manage your time and space wisely.**

Kindly ensure that all your final answers are written clearly in the designated table provided in the question paper. This table is mandatory, and answers not included in the table may not be graded.

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Write your ID number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

Section A

Write the correct choice (the most appropriate answer) in the table given below. [0.75x12=9]

Q #	1	2	3	4	5	6
Answer						
Q #	7	8	9	10	11	12
Answer						

- Which of the following is an objective of Computer Security?
 A) Confidentiality
 B) Integrity
 C) Availability
 D) All of these
- Which of the following is not a cybercrime?
 A) Denial of Service
 B) Man in the Middle
 C) Malware
 D) AES
- ECB in the context of digital cryptography stands for:
 A) Electrical Circuit Board
 B) Electronic Code Book
 C) Electrical Code Book
 D) Electronic Circuit Book
- Which term is used in computer security to protect your data from getting disclosed?
 A) Integrity
 B) Authentication
 C) Confidentiality
 D) Availability
- Which term is used in computer security to protect the data from being modified by the unauthorized user?
 A) Integrity
 B) Authentication
 C) Confidentiality
 D) Availability
- Which of the following is the other name of symmetric key cryptography?
 A) Private Key
 B) Secret Key
 C) Ideal Key
 D) Both A and B
- What is the key bits value of Triple DES?
 A) 112
 B) 116
 C) 118
 D) 121
- Which of the following is the structure of DES?
 A) Fiestal Network
 B) Luby Rackoff Block Cipher
 C) Assymetric Cipher
 D) Both A and B

9. Does the set of residue classes (mod 3) form a group with respect to modular addition
 - A) YES
 - B) NO
 - C) Cannot say
 - D) Insufficient Data
10. What is the gcd value of the pair (88 and 220) using Euclid algorithm
 - A) 22
 - B) 44
 - C) 11
 - D) 88
11. Which type of cipher can be categorized into two broad categories: mono-alphabetic and Poly-alphabetic.
 - A) Substitution
 - B) Transposition
 - C) either (A) or (B)
 - D) neither (A) nor (B)
12. Which of the following is the bit size of AES?
 - A) 128
 - B) 192
 - C) 256
 - D) All of these

Section B [21 Marks]

NOTE:

1. Write answers in the table below.
2. For each question, you must show all your work and steps leading to your final answer. Use the space provided beneath each question to demonstrate your calculations, reasoning, or explanations.
3. If you need additional space, use the extra sheets provided at the end of the question paper.

Question	Answer
1.	W8 =
2.	(a)
	(b)
3.	Cipher text= Show steps of Encryption and Decryption in provided space.
4.	(i) Public key of Dudu =
	(ii) $C_m =$
	(iii) Show steps in provided space

1. In the second round of the AES key expansion process, the values of the key schedule words :

w4=dc 90 37 b0

w5 = 9b 49 df e9

w6= 97 fe 72 3f

w7=38 81 15 a7 are given.

It is known that the value of RC [2] is 02. Additionally, the S-Box is provided (check Appendix page 10).

Calculate the value for w8.

[3 Marks]

Sol:

Rot Word	X2=
SubWord	Y2=
Rcon (2)	
$Y2 \oplus Rcon(2)$	
W8 =	

2. Consider performing a known-plaintext attack against the Data Encryption Standard (DES) using a single pair of plaintext and cipher text. **[2.5*2=5 Marks]**

(a) In the worst-case scenario, how many keys are required to be tested if an exhaustive key search is employed in a straightforward manner?

Rough Work:

(b) In the average-case scenario, how many keys need to be tested if exhaustive search is employed in a straightforward manner?

Rough Work:

3. Perform encryption and decryption using the RSA algorithm , following parameters are given for RSA:

(i) Prime numbers $p=3$ and $q=11$.

(ii) Public exponent $e=7$.

(iii) Plaintext (Message) $m=5$. **[7 Marks]**

Sol:

Encryption (Show Steps)

Decryption(Show Steps)

4. Using Elliptic Curve Cryptography over Elgamal Cryptosystem solve the following problem.

The cryptosystem parameters are:

Elliptic Curve: $E_{11}(1, 6)$

Base Point (generator) $G = (2, 7)$.

Secret key for Dudu is $n_d = 7$.

Bubu wishes to send encrypted message to Dudu.

Points on the curve are provided. (check Appendix page 10)

[2*3=6 Marks]

(i) Find public key of Dudu.

Rough Work:

- (ii) Bubu wishes to encrypt the message $P_m = (10, 9)$ and chooses the random value $k=3$. Determine the cipher text C_m .

Rough Work:

- (iii) Show the calculation by which Dudu recovers P_m from C_m .

Appendix

S-Box of Advanced Encryption Standard(AES)

		Y															
X		0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
	0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
	1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
	2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
	3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
	4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
	5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
	6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
	7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
	8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
	9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
	a	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
	b	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
	c	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
	d	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
	e	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
	f	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	BO	54	BB	16

Decimal Binary Hexadecimal Table

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Ponits on Elliptic Curve: $E_{11}(1, 6)$

{ $2G = (5, 2)$, $3G = (8, 3)$, $4G = (10, 2)$, $5G = (3, 6)$, $6G = (7, 9)$, $7G = (7, 2)$, $8G = (3, 5)$,
 $9G = (10, 9)$, $10G = (8, 8)$, $11G = (5, 9)$, $12G = (2, 4)$, $13G = (2, 7)$ }

ALL THE BEST☺

STUDENT ID :

